

Stony Brook Institute at
Anhui University (SBIAHU)

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Applied Calculus IV

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Week 3

- Review
- 2.2
- 2.3

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First-Order Differential Equations

- 2.1 Solution Curves Without a Solution
- 2.2 Separable Equations
- 2.3 Linear Equations
- 2.4 Exact Equations
- 2.5 Solutions by Substitutions
- 2.6 A Numerical Method

First-order ODEs

$$y' = f(x, y)$$

$$y' = g(x)h(y)$$

Sometimes,
variables are
separable



Separable Equation

Example:

$$\frac{dy}{dx} = y^2 x e^{3x+4y}$$



$$f(x, y) = y^2 x e^{3x+4y} = \overset{g(x)}{\downarrow} (x e^{3x}) \overset{h(y)}{\downarrow} (y^2 e^{4y}),$$

Exercises 2.2

1. $\frac{dy}{dx} = \sin 5x$

3. $dx + e^{3x} dy = 0$

5. $x \frac{dy}{dx} = 4y$

7. $\frac{dy}{dx} = e^{3x+2y}$

2. $\frac{dy}{dx} = (x + 1)^2$

4. $dy - (y - 1)^2 dx = 0$

6. $\frac{dy}{dx} + 2xy^2 = 0$

8. $e^x y \frac{dy}{dx} = e^{-y} + e^{-2x-y}$

Solving a separable first-order ODE

$$y' = g(x)h(y)$$

Divided by $h(y)$ $p(y)y' = g(x)$

$$p(y) = \frac{1}{h(y)}$$

$$p(y)dy = g(x)dx$$

$$\int p(y) dy = \int g(x)dx$$

$$\int p(\varphi(x))\varphi(x)dx = \int g(x)dx$$

$$y = \varphi(x)$$

Example: Solve $(1 + x)dy - ydx = 0$

$$(1 + x)dy = ydx$$

$$\frac{dy}{y} = \frac{dx}{1 + x}$$

$$\ln|y| = \ln|1 + x| + c_1$$

$$\ln|y| = \ln|1 + x| + \ln|c| = \ln |c(1 + x)|$$



$$y = c(1 + x)$$

Example: Solve $\frac{dy}{dx} = -\frac{x}{y}$, and $y(4) = -3$

$$ydy = -xdx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + c_1$$

$$y^2 = -x^2 + c$$

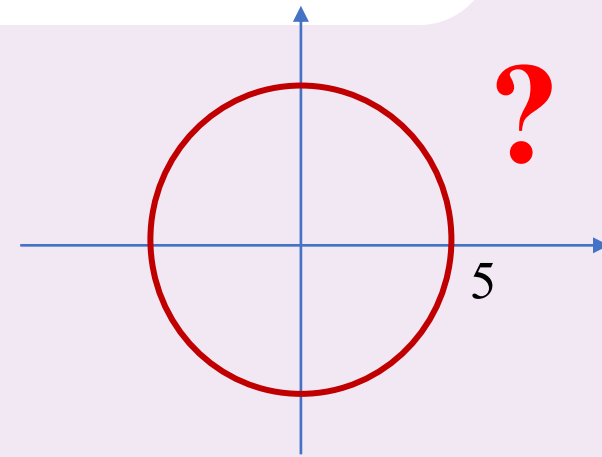
$$y^2 + x^2 = c$$

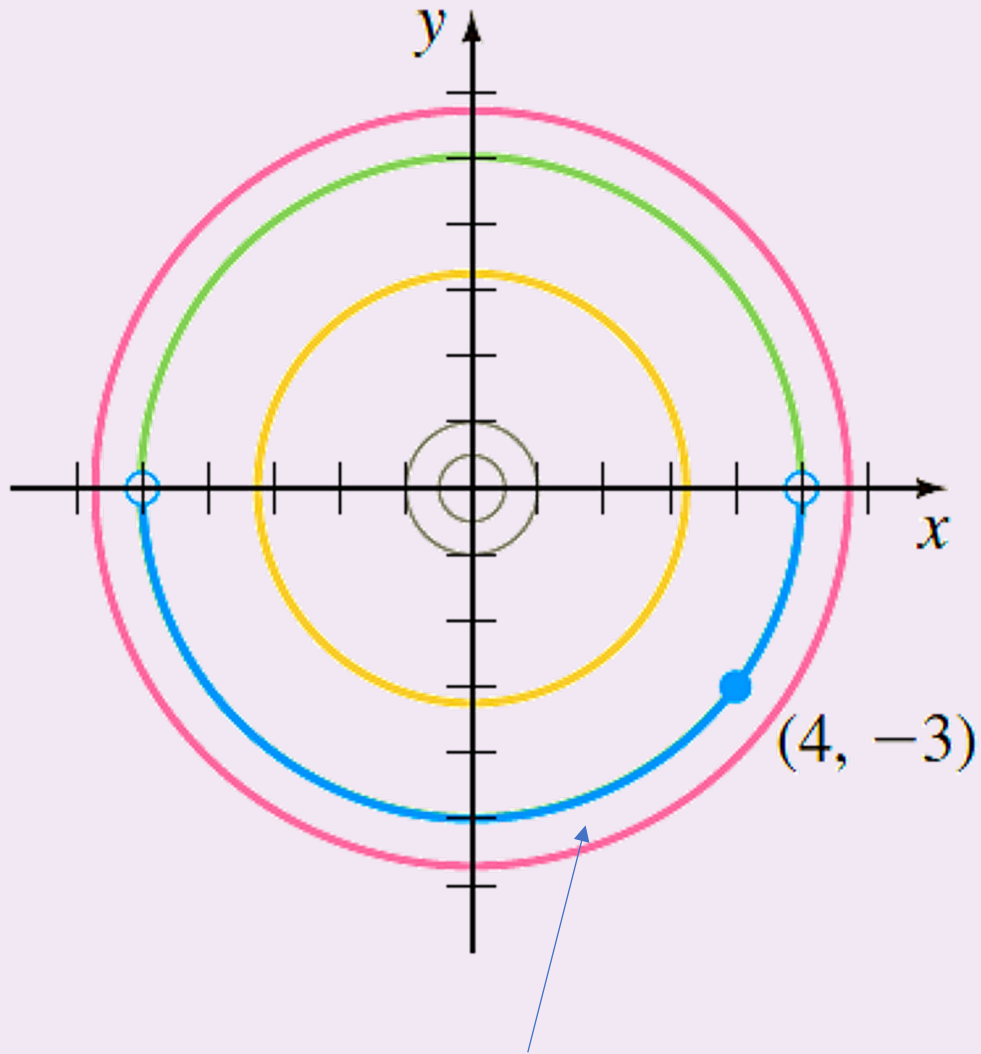
$$(-3)^2 + 4^2 = 25 = c$$

$$y^2 + x^2 = 25$$

➡ $y = -\sqrt{25 - x^2}, \quad -5 < x < 5$

Note: The solution must be a differentiable function
(half-circle is feasible but never a full circle)





Solution Curve
(blue one)

Note that in this general solution, $y^2 + x^2 = c$, each particular solution is a half-circle, **not** a full circle

$$y^2 + x^2 = c \quad \sim \quad \begin{cases} y = +\sqrt{c - x^2}, \\ y = -\sqrt{c - x^2}, \end{cases}$$

Example: Find the solution of IVP

$$\frac{dy}{dx} = -\frac{x}{y}, \quad \text{and } y(4) = 3$$

Solution: The solution curve is

$$y = +\sqrt{25 - x^2}, \quad -5 < x < 5$$

(The green curve in this figure)

Example: Solve the following initial value problem

$$(e^{2y} - y)\cos x \frac{dy}{dx} = e^y \sin 2x,$$

$$y(0) = 0$$

Example: To solve $(e^{2y} - y)\cos x \frac{dy}{dx} = e^y \sin 2x$, $y(0) = 0$:

$$\frac{(e^{2y} - y)}{e^y} dy = \frac{\sin 2x}{\cos x} dx$$

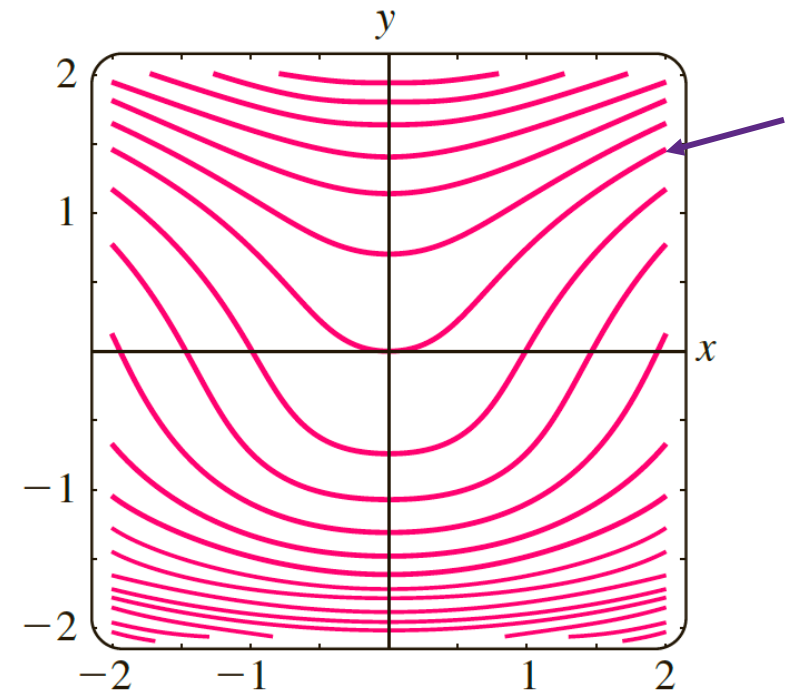
$$(e^y - ye^{-y})dy = 2\sin x dx$$

$$e^y + ye^{-y} + e^{-y} = -2\cos x + c$$

$$1 + 0 + 1 = -2 + c$$



$$e^y + ye^{-y} + e^{-y} = 4 - 2\cos x$$



In Problems 23–28 find an explicit solution of the given initial-value problem.

→ 23. $\frac{dx}{dt} = 4(x^2 + 1), \quad x(\pi/4) = 1$

24. $\frac{dy}{dx} = \frac{y^2 - 1}{x^2 - 1}, \quad y(2) = 2$

→ 25. $x^2 \frac{dy}{dx} = y - xy, \quad y(-1) = -1$

26. $\frac{dy}{dt} + 2y = 1, \quad y(0) = \frac{5}{2}$

→ 27. $\sqrt{1 - y^2} dx - \sqrt{1 - x^2} dy = 0, \quad y(0) = \frac{\sqrt{3}}{2}$

28. $(1 + x^4) dy + x(1 + 4y^2) dx = 0, \quad y(1) = 0$

23. $\frac{dx}{dt} = 4(x^2 + 1), \quad x(\pi/4) = 1$

Integrating:

$$\frac{1}{x^2+1} dx = 4dt$$

$$\Rightarrow \int \frac{1}{x^2+1} dx = \int 4dt$$

$$\Rightarrow \tan^{-1} x = 4t + c$$

$$x\left(\frac{\pi}{4}\right) = 1$$

$$\tan^{-1} 1 = 4 \cdot \frac{\pi}{4} + c$$

$$\frac{\pi}{4} = \pi + c$$

$$\Rightarrow c = -\frac{3\pi}{4}$$

Explicit solution

$$\tan^{-1} x = 4t - \frac{3\pi}{4}$$

$$\Rightarrow x = \tan\left(4t - \frac{3\pi}{4}\right)$$

$$25. \ x^2 \frac{dy}{dx} = y - xy, \quad y(-1) = -1$$

Separating variables and integrating:

$$\frac{1}{y} dy = \frac{1-x}{x^2} dx$$

$$\frac{1}{y} dy = \left(\frac{1}{x^2} - \frac{1}{x} \right) dx$$

$$\Rightarrow \ln |y| = -\frac{1}{x} - \ln |x| + c$$

$$\Rightarrow xy = c_1 e^{-1/x}$$

$$y(-1) = -1$$

$$\Rightarrow (-1)(-1) = c_1 e^{-1/(-1)}$$

$$\Rightarrow 1 = c_1 e^1$$

$$\Rightarrow c_1 = e^{-1}$$

$$\Rightarrow xy = e^{-1} \cdot e^{-1/x}$$

$$\Rightarrow xy = e^{-1-1/x}$$

$$\Rightarrow y = \frac{e^{-(1+1/x)}}{x}$$

$$27. \sqrt{1-y^2} dx - \sqrt{1-x^2} dy = 0, \quad y(0) = \frac{\sqrt{3}}{2}$$

Separating variables and integrating:

$$\frac{dx}{\sqrt{1-x^2}} - \frac{dy}{\sqrt{1-y^2}} = 0$$

$$\Rightarrow \sin^{-1} x - \sin^{-1} y = c$$

$$y(0) = \frac{\sqrt{3}}{2}$$

$$\sin^{-1} 0 - \sin^{-1}(\sqrt{3}/2) = c$$

$$\Rightarrow 0 - \pi/3 = c$$

$$\Rightarrow c = -\pi/3$$

Implicit solution

$$\sin^{-1} x - \sin^{-1} y = \pi/3$$

Explicit solution by using formula from trigonometry

$$y = \sin(\sin^{-1} x + \pi/3)$$

$$= x \cos(\pi/3) + \sqrt{1-x^2} \sin(\pi/3)$$

$$= \frac{x}{2} + \frac{\sqrt{3}\sqrt{1-x^2}}{2}$$

In Problems 31–34 find an explicit solution of the given initial-value problem. Determine the exact interval I of definition by analytical methods. Use a graphing utility to plot the graph of the solution.

31. $\frac{dy}{dx} = \frac{2x + 1}{2y}, \quad y(-2) = -1$

32. $(2y - 2) \frac{dy}{dx} = 3x^2 + 4x + 2, \quad y(1) = -2$

33. $e^y dx - e^{-x} dy = 0, \quad y(0) = 0$

34. $\sin x dx + y dy = 0, \quad y(0) = 1$

34. Solution

$$\sin x \, dx + y \, dy = 0$$

$$\Rightarrow \int \sin x \, dx + \int y \, dy = \int 0 \, dx$$

$$\Rightarrow -\cos x + \frac{1}{2}y^2 = c$$

$$-\cos 0 + \frac{1}{2}(1)^2 = c$$

$$-1 + \frac{1}{2} = c$$

$$\Rightarrow c = -\frac{1}{2}$$

$$-\cos x + \frac{1}{2}y^2 = -\frac{1}{2}$$

$$\Rightarrow \frac{1}{2}y^2 = \cos x - \frac{1}{2}$$

$$\Rightarrow y^2 = 2\cos x - 1$$

Implicit solution

?

Considering initial condition $y(0) = 1 > 0$

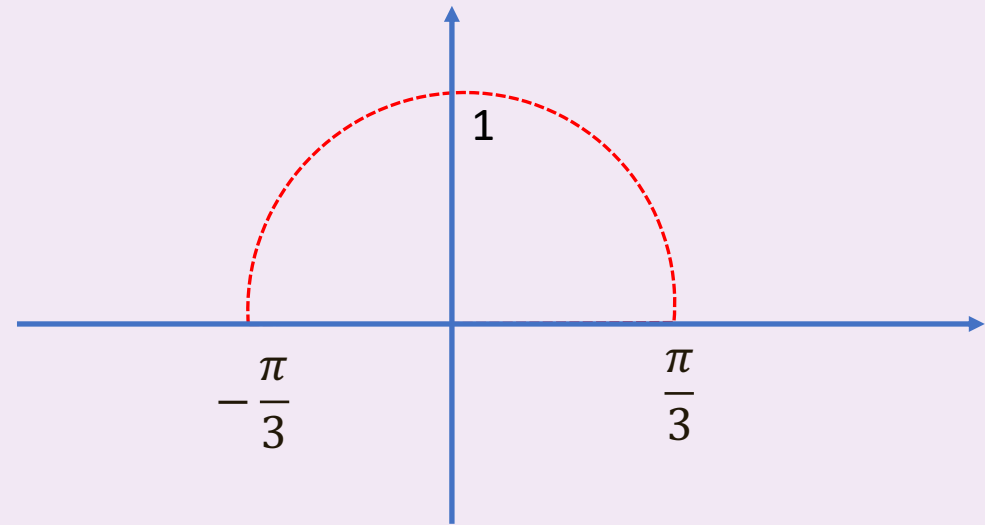
$$y = \sqrt{2 \cos x - 1}$$

$$2 \cos x - 1 > 0$$

$$\cos x > \frac{1}{2}$$

$$\Rightarrow -\frac{\pi}{3} < x < \frac{\pi}{3}$$

Interval of Definition



Note that

$$y^2 = 2 \cos x + c$$

$\sim$

$$y = +\sqrt{2 \cos x + c} \quad (a)$$

$$y = -\sqrt{2 \cos x + c} \quad (b)$$

and each particular solution (obtained by different values of parameter c),
satisfies either (a) or (b)

36. Find a solution of $x \frac{dy}{dx} = y^2 - y$ that passes through the indicated points.

- (a) $(0, 1)$ (b) $(0, 0)$ (c) $(\frac{1}{2}, \frac{1}{2})$ (d) $(2, \frac{1}{4})$

For interested students:

This exercise for initial condition $y(0) = 1$, and also the example explained in page 51 show that the solution of initial value problems might not be unique!

Solution

$$\frac{dy}{y^2 - y} = \frac{dx}{x}$$

$$\Rightarrow \int \frac{dy}{y(y-1)} = \ln |x| + c$$

$$\int \left(\frac{1}{y-1} - \frac{1}{y} \right) dy = \ln |x| + c$$

$$\Rightarrow \ln |y-1| - \ln |y| = \ln |x| + c$$

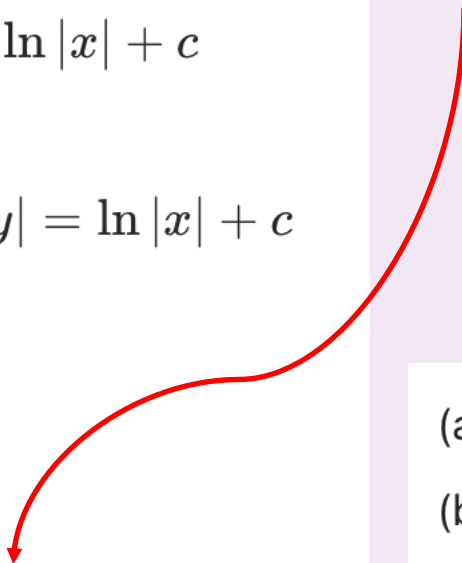
$$\Rightarrow \ln \left| \frac{y-1}{y} \right| = c$$

$$\Rightarrow \frac{y-1}{y} = e^c = c_1$$

$$y = \frac{1}{1-c_1 x}$$

Note that $y(x) = 0$ is a singular solution of the DE

It is not a unique solution!

- 
- 
- (a) $x = 0$ and $y = 1$: $1 = 1/(1 - 0)$, true for all c_1 ; $y = 1/(1 - c_1 x)$.
 - (b) $x = 0$ and $y = 0$ in $y = 1/(1 - c_1 x)$: $0 = 1$; $y = 0$.
 - (c) $x = \frac{1}{2}$ and $y = \frac{1}{2}$: $\frac{1}{2} = 1/(1 - \frac{1}{2}c_1)$, $c_1 = -2$, $y = 1/(1 + 2x)$.
 - (d) $x = 2$ and $y = \frac{1}{4}$: $\frac{1}{4} = 1/(1 - 2c_1)$, $c_1 = -\frac{3}{2}$, $y = 2/(2 + 3x)$.

First-Order Differential Equations

- 2.1 Solution Curves Without a Solution
- 2.2 Separable Equations
- 2.3 Linear Equations
- 2.4 Exact Equations
- 2.5 Solutions by Substitutions
- 2.6 A Numerical Method

DEFINITION 2.3.1 Linear Equation

A first-order differential equation of the form

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x), \quad (1)$$

is said to be a **linear equation** in the variable y .

$$\frac{dy}{dx} + P(x)y = f(x)$$

The standard form

$$\mu(x) = e^{\int P(x) dx}$$

Technique (multiplying the equation by this term)

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

Because

$$e^{\int P(x)dx} \frac{dy}{dx} + P(x)e^{\int P(x)dx} y = e^{\int P(x)dx} f(x)$$

$$\frac{d}{dx} \left[e^{\int P(x)dx} y \right] = e^{\int P(x)dx} f(x)$$

$$e^{\int P(x)dx} y = \int e^{\int P(x)dx} f(x) dx + c$$

$$y = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx + ce^{-\int P(x)dx}$$

Example: Solve $\frac{dy}{dx} - 3y = 0$

$$P(x) = -3 \quad \rightarrow \quad e^{\int (-3) dx} = e^{-3x}$$

$$e^{-3x} \frac{dy}{dx} - 3e^{-3x} y = 0$$

$$\frac{d}{dx} [e^{-3x} y] = 0$$

$$e^{-3x} y = c \quad \rightarrow \quad y = ce^{3x}$$

Example: Solve $x \frac{dy}{dx} - 4y = x^6 e^x$

$$\frac{dy}{dx} - \frac{4}{x}y = x^5 e^x$$

$$P(x) = -\frac{4}{x} \quad \rightarrow \quad e^{-\int \frac{4}{x} dx} = x^{-4} \quad -\int \frac{4}{x} dx = -4 \ln x = \ln x^{-4}$$

$$x^{-4} \frac{dy}{dx} - 4x^{-5}y = x e^x$$

$$\frac{d}{dx} [x^{-4}y] = x e^x$$

$$x^{-4}y = x e^x - e^x + c \Rightarrow y = x^5 e^x - x^4 e^x + c x^4$$

Step by Step

- Change it to the standard form $\frac{dy}{dx} + P(x)y = f(x)$
- Identify $P(x)$
- Find the integrating factor, $e^{\int P(x) dx}$
- Multiply the equation by the integrating factor
- Integrate both sides

Example: Find the solution of $\frac{dy}{dx} - y = \sin x$

$$P(x) = -1$$

$$e^{-\int dx} = e^{-x}$$

$$e^{-x} \frac{dy}{dx} - e^{-x} y = e^{-x} \sin x \rightarrow [e^{-x} y] = e^{-x} \sin x$$

$$e^{-x} y = -\frac{1}{2} e^{-x} (\sin x + \cos x) + c$$

$$y = -\frac{1}{2} (\sin x + \cos x) + ce^x$$

Identify $P(x)$

Find the integrating factor

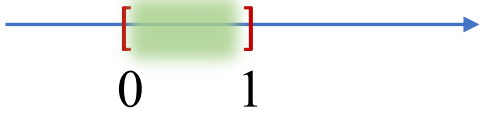
Multiply by IF

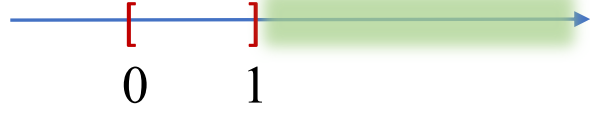
Integrate both sides
(use the integration table)



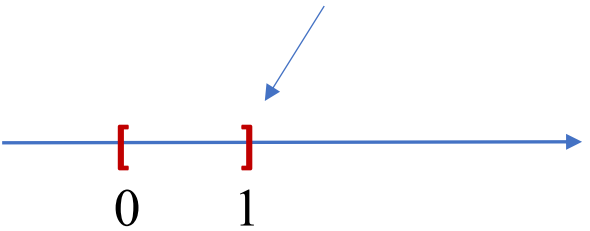
Example: Solve $y' + y = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$, where $y(0) = 0$

$$P(x) = 1 \rightarrow e^{\int dx} = e^x$$

$$e^x \frac{dy}{dx} + e^x y = e^x \rightarrow \frac{d}{dx} [e^x y] = e^x \rightarrow y = e^{-x}(e^x + c_1) = 1 + e^{-x}c_1$$


$$e^x \frac{dy}{dx} + e^x y = 0 \rightarrow \frac{d}{dx} [e^x y] = 0 \rightarrow y = c_2 e^{-x}$$


$$y(0) = 0 \rightarrow 0 = 1 + c_1 \rightarrow c_1 = -1 \rightarrow y = 1 - e^{-x}$$

$$y(1) = y(1^-) = y(1^+) \rightarrow 1 - e^{-1} = c_2 e^{-1} \rightarrow c_2 = e - 1$$


➡
$$y = \begin{cases} 1 - e^{-x} & 0 \leq x \leq 1 \\ (1 - e)e^{-x} & x > 1 \end{cases}$$

Exercises 2.3

A **transient term** is a component of a solution that fades and becomes negligible (It describes state or things that lasts for only a short time)

In Problems 1–24 find the general solution of the given differential equation. Give the largest interval I over which the general solution is defined. Determine whether there are any transient terms in the general solution.

1. $\frac{dy}{dx} = 5y$

2. $\frac{dy}{dx} + 2y = 0$

3. $\frac{dy}{dx} + y = e^{3x}$

4. $3 \frac{dy}{dx} + 12y = 4$

5. $y' + 3x^2y = x^2$

6. $y' + 2xy = x^3$

7. $x^2y' + xy = 1$

8. $y' = 2y + x^2 + 5$

9. $x \frac{dy}{dx} - y = x^2 \sin x$

10. $x \frac{dy}{dx} + 2y = 3$

11. $x \frac{dy}{dx} + 4y = x^3 - x$

12. $(1 + x) \frac{dy}{dx} - xy = x + x^2$

13. $x^2y' + x(x + 2)y = e^x$

14. $xy' + (1 + x)y = e^{-x} \sin 2x$

15. $y \, dx - 4(x + y^6) \, dy = 0$

3. Solution $\frac{dy}{dx} + y = e^{3x}$

$$P(x) = 1 \rightarrow e^{\int dx} = e^x$$

$$e^x \frac{dy}{dx} + e^x y = e^{4x}$$

$$\frac{d}{dx} [e^x y] = e^{4x}$$

$$e^x y = \frac{1}{4} e^{4x} + c$$

The transient
term is ce^{-x}


$$y = \frac{1}{4} e^{3x} + ce^{-x}$$

6. Solution $\frac{dy}{dx} + 2xy = x^3$

$$P(x) = 2x \rightarrow e^{\int 2x dx} = e^{x^2}$$

$$e^{x^2} \frac{dy}{dx} + 2e^{x^2} xy = x^3 e^{x^2}$$

$$\frac{d}{dx} [e^{x^2} y] = x^3 e^{x^2}$$

$$e^{x^2} y = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + c$$

The transient
term is ce^{-x^2}


$$y = \frac{1}{2} x^2 - \frac{1}{2} + ce^{-x^2}$$

Using integration by parts $\int u \, dv = uv - \int v \, du$

$$\int x^3 e^{x^2} \, dx = \int \textcolor{red}{x^2} \textcolor{blue}{x} e^{x^2} \, dx = \frac{1}{2} x^2 e^{x^2} - \int x e^{x^2} \, dx = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2}$$

$$\textcolor{red}{u = x^2 \rightarrow du = 2x \, dx}$$

$$\textcolor{blue}{x e^{x^2} \, dx = dv \rightarrow v = \frac{1}{2} e^{x^2}}$$

In Problems 25–36 solve the given initial-value problem. Give the largest interval I over which the solution is defined.

25. $\frac{dy}{dx} = x + 5y, \quad y(0) = 3$

26. $\frac{dy}{dx} = 2x - 3y, \quad y(0) = \frac{1}{3}$

27. $xy' + y = e^x, \quad y(1) = 2$

28. $y \frac{dx}{dy} - x = 2y^2, \quad y(1) = 5$

25. Solution

$$\frac{dy}{dx} - 5y = x \quad \text{now, it is in the right form}$$

$$P(x) = -5 \rightarrow e^{-\int 5dx} = e^{-5x}$$

$$e^{-5x} \frac{dy}{dx} - 5e^{-5x}y = xe^{-5x}$$

$$\frac{d}{dx} [e^{-5x}y] = xe^{-5x}$$

$$e^{-5x}y = -\frac{1}{5}xe^{-5x} - \frac{1}{25}e^{-5x} + c$$

$$y = -\frac{1}{5}x - \frac{1}{25} + ce^{-5x}$$



$$I = (-\infty, \infty)$$

27. Solution

$$y' + \frac{1}{x}y = \frac{1}{x}e^x$$

$$e^{\int \frac{1}{x} dx} = x$$

$$\frac{d}{dx}[xy] = e^x$$

$$y = \frac{1}{x}e^x + \frac{c}{x}$$

$$2 = \frac{1}{1}e^1 + \frac{c}{1} \implies c = 2 - e$$

$$y = \frac{1}{x}e^x + \frac{2-e}{x}$$

As we can see, $y(x) = \frac{1}{x}e^x + \frac{2-e}{x}$ is not defined at $x = 0$.

The suggested interval of definition cannot include this point.

The largest interval of definition can be

$$I = (-\infty, 0)$$

or

$$I = (0, \infty)$$

28. Solution

$$y \frac{dx}{dy} - x = 2y^2$$

$$\frac{dx}{dy} - \frac{1}{y}x = 2y$$

$$\mu(y) = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

$$\frac{1}{y} \frac{dx}{dy} - \frac{1}{y^2} x = 2$$

$$\frac{d}{dy} \left[\frac{x}{y} \right] = 2$$

$$x = 2y^2 + cy$$

$$y(1) = 5 \implies c = -\frac{49}{5}$$

$$x = 2y^2 - \frac{49}{5}y \quad 0 < y$$

The solution $x(y)$ is defined on

$$I = (0, \infty)$$

Remark: In this differential equation, y is the **independent** variable. Sometimes it can be used as a trick to swap the role of x and y

39. $\frac{dy}{dx} + 2xy = f(x), \quad y(0) = 2$, where

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$$

39. Solution

$$P(x) = 2x \rightarrow e^{\int 2x dx} = e^{x^2}$$

$$e^{x^2} \frac{dy}{dx} + 2e^{x^2} xy = \begin{cases} xe^{x^2} & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

$$\frac{d}{dx} [e^{x^2} y] = \begin{cases} xe^{x^2} & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

$$e^{x^2} y = \begin{cases} \frac{1}{2} e^{x^2} + c_1 & 0 \leq x \leq 1 \\ c_2 & x > 1 \end{cases}$$

$$y(x) = \begin{cases} \frac{1}{2} + c_1 e^{-x^2} & 0 \leq x \leq 1 \\ c_2 e^{-x^2} & x > 1 \end{cases}$$

Using the initial condition and differentiability of $y(x)$:

$$y(0) = 2 \rightarrow 2 = \frac{1}{2} + c_1 \rightarrow c_1 = \frac{3}{2}$$

$$y(1^-) = y(1^+) \rightarrow \frac{1}{2} + \frac{3}{2} e^{-1} = c_2 e^{-1} \rightarrow c_2 = \frac{1}{2} e + \frac{3}{2}$$

$$\rightarrow y(x) = \begin{cases} \frac{1}{2} + \frac{3}{2} e^{-x^2} & 0 \leq x \leq 1 \\ (\frac{1}{2} e + \frac{3}{2}) e^{-x^2} & x > 1 \end{cases}$$

41. $\frac{dy}{dx} + P(x)y = 4x, \quad y(0) = 3, \text{ where}$

$$P(x) = \begin{cases} 2, & 0 \leq x \leq 1 \\ -\frac{2}{x}, & x > 1 \end{cases}$$

41. Solution

For the interval $[0, 1]$:

- Differential equation: $y' + 2y = 4x$
- Integrating factor: $e^{\int 2dx} = e^{2x}$
- Transformed equation: $\frac{d}{dx}[e^{2x}y] = 4xe^{2x}$
- Integral result: $e^{2x}y = \int 4xe^{2x}dx = 2xe^{2x} - e^{2x} + c_1$
- General solution: $y = 2x - 1 + c_1e^{-2x}$
- Using initial condition $y(0) = 3$: $3 = -1 + c_1 \Rightarrow c_1 = 4$
- Particular solution: $y = 2x - 1 + 4e^{-2x}$

For the interval $(1, \infty)$:

- Differential equation: $y' - \frac{2}{x}y = 4x$
- Integrating factor: $e^{\int (-\frac{2}{x})dx} = e^{-2\ln x} = x^{-2}$
- Transformed equation: $\frac{d}{dx}[x^{-2}y] = \frac{4}{x}$
- Integral result: $x^{-2}y = \int \frac{4}{x}dx = 4\ln x + c_2$
- General solution: $y = 4x^2 \ln x + c_2x^2$



$$y(x) = \begin{cases} 2x - 1 + 4e^{-2x}, & 0 \leq x \leq 1 \\ 4x^2 \ln x + c_2x^2, & x > 1 \end{cases}$$

To find parameter c_2 :

The solution must be differentiable at $x = 1$, which means $y(1) = y(1^-) = y(1^+)$

$$y(1^-) = y(1^+) \rightarrow 2 - 1 + 4e^{-2} = 0 + c_2 \rightarrow c_2 = 1 + 4e^{-2}$$


$$y = \begin{cases} 2x - 1 + 4e^{-2x}, & 0 \leq x \leq 1 \\ 4x^2 \ln x + (1 + 4e^{-2})x^2, & x > 1 \end{cases}$$

Basic

Integral

Formulas

$$1. \int u^n du = \frac{u^{n+1}}{n+1} + C \text{ (for } n \neq -1)$$

$$2. \int \frac{1}{u} du = \ln |u| + C$$

$$3. \int du = u + C$$

$$4. \int e^u du = e^u + C$$

$$5. \int a^u du = \frac{a^u}{\ln a} + C \text{ (for } a > 0, a \neq 1)$$

$$6. \int \ln u du = u \ln u - u + C$$

$$7. \int \sin u du = -\cos u + C$$

$$8. \int \cos u du = \sin u + C$$

$$9. \int \sin^2 u du = \frac{u}{2} - \frac{\sin 2u}{4} + C$$

$$10. \int \cos^2 u du = \frac{u}{2} + \frac{\sin 2u}{4} + C$$

$$11. \int \tan u du = -\ln |\cos u| + C = \ln |\sec u| + C$$

$$12. \int \sec u du = \ln |\sec u + \tan u| + C$$

$$13. \int \sec^2 u du = \tan u + C$$

$$14. \int \sec u \tan u du = \sec u + C$$

$$15. \int \cot u du = \ln |\sin u| + C = -\ln |\csc u| + C$$

$$16. \int \csc u du = \ln |\csc u - \cot u| + C$$